

MOIKIPAA, Finland

WMO: 029210

Lat: 62.888N

Lon: 21.091E

Elev: 13

StdP: 101.17

Time Zone: 2.00 (EUE)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.7	-21.3	-27.3	0.3	-24.5	-23.9	0.4	-21.2	18.9	-5.8	16.0	-2.4	1.8	40	0.817

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.0	20.8	17.3	19.4	16.3	18.2	15.5	17.8	19.8	16.9	18.9	16.0	17.8	4.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.1	19.0	15.9	11.3	18.1	15.1	10.7	17.2	50.4	19.9	47.4	19.0	45.0	17.9	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.6	13.1	11.8	DB	-24.1	23.5	4.2	2.4	-27.1	25.2	-29.6	26.7	-31.9	28.1	-35.0	29.8	(4)
(5)			WB	-24.3	19.2	4.1	1.6	-27.3	20.4	-29.7	21.3	-32.0	22.2	-35.0	23.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-7.7	-6.8	-3.5	1.0	7.1	12.3	15.5	14.3	10.3	6.1	0.8	-4.0		
	DBStd	8.98	7.63	6.47	4.51	3.36	3.25	2.95	2.09	2.34	2.36	2.58	4.31	5.16		
	HDD10.0	2676	547	470	417	271	101	10	0	1	24	122	277	436		
	HDD18.3	5294	806	703	676	521	349	181	92	126	241	379	527	694		
	CDD10.0	429	0	0	0	0	11	79	171	134	33	1	0	0		
	CDD18.3	7	0	0	0	0	0	1	4	1	0	0	0	0		
	CDH23.3	9	0	0	0	0	0	1	8	1	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	5.9	6.1	6.0	5.9	5.6	5.3	5.1	4.8	5.3	6.0	7.0	7.2	6.5		
(15) Precipitation	PrecAvg	557	45	32	32	23	35	51	65	54	51	62	56	53		
	PrecMax	702	77	67	54	41	75	104	106	102	110	111	129	98		
	PrecMin	437	10	0	11	2	5	12	11	9	19	18	11	17		
	PrecStd	80	20	18	13	11	22	23	30	30	27	26	27	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	3.2	3.4	12.6	18.7	21.8	24.2	21.4	17.3	12.3	8.0	4.2		
		MCWB	2.0	2.2	2.2	9.0	14.9	17.3	20.4	17.7	13.9	10.9	7.1	3.2		
	2%	DB	2.2	2.3	2.7	9.1	15.4	19.6	21.3	19.7	15.3	11.0	7.1	3.5		
		MCWB	1.4	1.6	1.6	6.7	12.7	16.0	18.1	16.6	13.4	9.8	6.1	2.7		
	5%	DB	1.5	1.6	2.1	6.7	13.1	18.0	19.9	18.6	14.3	10.2	6.4	2.9		
		MCWB	0.8	1.1	1.2	5.0	10.3	14.7	16.7	16.2	12.7	9.3	5.5	2.2		
	10%	DB	0.8	0.9	1.3	5.2	11.4	16.7	18.7	17.5	13.4	9.5	5.5	2.2		
		MCWB	0.3	0.4	0.6	3.9	9.0	13.8	15.8	15.5	11.9	8.6	4.6	1.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.5	2.5	9.4	15.4	17.7	21.3	18.6	14.8	11.1	7.3	3.6		
		MCDB	2.9	2.8	3.0	11.7	18.7	20.7	23.5	20.4	16.6	11.8	7.6	3.9		
	2%	WB	1.6	1.8	1.9	6.7	13.1	16.6	18.5	17.6	13.8	10.2	6.4	2.8		
		MCDB	1.9	2.2	2.4	8.7	15.4	19.1	21.1	19.1	14.8	10.8	7.0	3.2		
	5%	WB	0.9	1.1	1.4	5.3	10.5	15.2	17.1	16.6	13.1	9.5	5.7	2.3		
		MCDB	1.3	1.5	1.8	6.5	12.5	17.5	19.3	18.1	14.0	10.1	6.3	2.8		
	10%	WB	0.3	0.5	0.7	4.1	9.3	14.1	16.3	15.7	12.3	8.8	4.9	1.6		
		MCDB	0.8	0.8	1.2	5.2	11.0	16.1	18.2	17.2	13.2	9.4	5.5	2.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.3	4.8	4.4	4.5	4.1	4.0	3.5	3.1	2.7	3.0	4.5		
		MCDBR	3.3	2.9	3.5	6.1	6.5	5.7	5.8	5.0	4.2	2.9	2.5	2.7		
		MCWBR	3.4	2.8	2.9	4.2	3.8	3.2	3.5	3.4	3.2	2.5	2.5	2.6		
		MCDBR	3.3	2.6	2.9	5.9	5.2	5.5	5.1	4.6	3.9	2.8	2.4	2.7		
	5% WB	MCWBR	3.4	2.6	2.5	4.3	3.6	3.3	3.5	3.5	3.3	2.5	2.5	2.7		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.298	0.318	0.358	0.356	0.352	0.374	0.362	0.336	0.319	0.387	0.350		
	taud		2.079	2.201	2.281	2.324	2.439	2.485	2.448	2.472	2.524	2.387	2.686	2.550		
	Ebn at Noon		393	636	781	824	861	871	838	817	771	618	363	221		
	Edh at Noon		32	63	85	101	98	96	97	87	69	53	29	19		
(44) All-Sky Solar Radiation	RadAvg		0.15	0.71	2.22	3.83	5.35	5.99	5.46	4.18	2.47	0.95	0.23	0.08		
	RadStd		0.02	0.14	0.26	0.40	0.50	0.41	0.46	0.36	0.23	0.18	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (44 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page